

A Trust Score calculation approach for Zero Trust Access System

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Abstract—Zero trust is a cybersecurity approach that focuses on cautious and continuous verification. Traditional trust presumptions are discarded in a zero trust scenario, and continual verification becomes necessary. This paper introduces a weighted average-based trust evaluation method specifically designed for zero trust environments. The approach incorporates multiple factors and assigns appropriate weights to these factors and calculates a trust score. A user or device is allowed access to a resource only if the calculated trust score is above a predefined threshold value. This method guarantees a comprehensive assessment of users and devices prior to granting them access to sensitive resources, and any weaknesses will result in a decrease in their trust score. The weighted average technique enables organizations to customize the trust evaluation process to their specific zero trust requirements through its capacity for openness, adaptability, and customization. The methodology offers a useful and effective way to evaluate trust in a dynamic and constantly changing security environment.

Index Terms—zero trust, trust score, security, weighted average

I. INTRODUCTION

The Firewalls, Intrusion Detection and Prevention Systems (IDPS), Unified Threat Management Systems (UTM) etc., are security devices that were developed to build a network security perimeter. These perimeter devices are designed to allow trusted users, applications and traffic inside while blocking out untrusted users [1]. The perimeter-based security approach considers that all the devices, users and applications inside the perimeter are trustworthy and all the entities outside the perimeter are untrusted. Hence, it focuses the traffic/input coming from outside the perimeter for inspection to identify the attacks. However, the myth of a defensive perimeter is challenged by the sharp increase in breaches caused by insider threats, compromised credentials, vulnerable devices and phishing attempts. Once an object has been authenticated, it is trusted for a very long time in this perimeter-based security model and architecture. As a result, once a harmful item has been authenticated, it can keep attacking and destroying the internal network [2]. As a result of this scenario, Zero Trust Architecture (ZTA) —which implies that neither internal nor

external entities are trustworthy—has become more popular. In the zero-trust concept, trust is never given implicitly but is instead constantly assessed. In 2010 [3], when the concept's basic model was revealed, an analyst at Forrester Research Inc. coined the phrase 'Zero Trust'. Zero Trust security was adopted by Google in their network in 2014 [4]–[7]. To make contextual access decisions, the Continuous Adaptive Risk and Trust Assessment (CARTA) framework, developed by Gartner, recommends continuously assessing all users or devices [8]. When it became crucial to safeguard the workplace following COVID-19 because employees were now working remotely, the Zero Trust idea started to gain rapid adoption. NIST released SP 800-207 Zero Trust Architecture Document in 2020 to standardize the technology [9]. It decreases the attack surface, lessens the impact and severity of cyberattacks, and shortens the time and expense required for a security breach response and recovery. The pictorial representation of the evolution of Zero Trust system is shown in Fig. 1.



Fig. 1. ZTA Timeline.

A. Zero Trust Architecture

The Zero Trust Architecture proposed by NIST [9] is shown in Fig 2. The Policy Decision Point (PDP), which is at the centre of this architecture, decides whether a user or device should be given access to the Zero Trust system. While the Policy Administrator (PA) oversees and maintains the policies for allowing and denying access, the PDP communicates with the Policy Enforcement Point (PEP) to implement them. The Policy Enforcement Point (PEP) is responsible for enabling, monitoring, and eventually terminating connections between a user and a resource.

Authentication of the user/device is carried out by the Identity Access Management (IAM) Module. Then, using

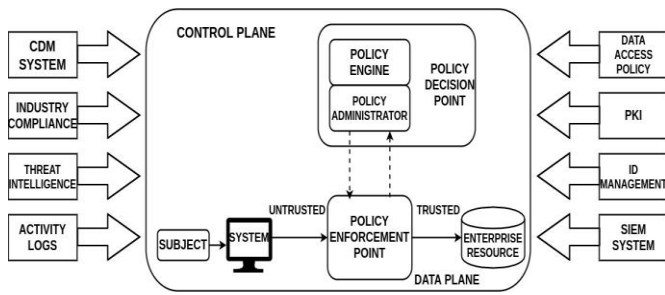


Fig. 2. NIST Zero Trust Architecture [9]

inputs from a wide range of data sources like device/user history, access log etc., a trust score is calculated for the user/device. The Trust Evaluation system continuously evaluates the trustworthiness of individuals, devices, and networks.

Together with other elements such as the Continuous Diagnostics and Mitigation (CDM) system, Enterprise Public Key Infrastructure (PKI), and SIEM (Security Information and Event Management) system, the Trust Evaluation component is a crucial part of the Zero Trust Architecture, offering a strong and flexible approach to security.

The trust score value, calculated by the trust evaluation system, is compared against a predefined threshold value and if the value is above the threshold, the request made by a user/device is granted or else denied. The computed trust score value helps us in determining whether we should trust an individual or device and hence trust score evaluation is a vital component of zero trust architecture.

In this paper, we propose a trust score calculation methodology using a weighted average algorithm that aggregates multiple criteria or factors and calculates an overall score. This method provides transparency, control, and flexibility in determining trustworthiness.

The organization of the paper is as follows: Section 1 describes the introduction and Section II gives the background of various trust evaluation methods. Section III explains the proposed system in detail. Section IV describes a sample scenario for trust score calculation. Finally, the conclusion as section V ends the paper.

II. BACKGROUND AND RELATED WORK

Trust is a critical component of network security which indicates the reliability or genuineness of a person or device. Without trust, it is impossible to establish secure communications and exchange data. You develop a connection of trust when you have enough faith in the qualities that an entity offers. ITU-T X.509, Section 3.3.54 [10] states that an entity can be considered to ‘trust’ a second entity when it assumes the second entity will act in the manner that the first entity anticipates. In [11], Trust Modelling for Security Architecture development is considered and Trust is stated as the ability to have confidence that something will or won’t happen in the manner that has been predicted or promised. Identification, authentication, accountability, authorization, and availability

all contribute to the enabling of confidence. Authors state that risk analysis and threat profiles are strongly intertwined and they also presented different trust models.

In [12] Shaikh et al. introduced a trust model that computes static and dynamic trust for assessing the security level of cloud computing services. A trust model that accumulates both positive, as well as negative evaluations about a particular member, is described by Josang et al. [13]. The model is based on the Bayesian network model and computes a member’s projected future behaviour using the beta probability density function. A trust model was established by E D Canedo et al. in [14], to rate reliable nodes and allow peer-to-peer secure file sharing in a private cloud. The authors in [15] suggested a dynamic trust updating technique. For the direct trust, time and transaction amount are considered. For reputation modelling, feedback similarity and weight were employed.

A distributed trust model based on the Bayesian Network is proposed in [16] to evaluate the trust of the virtual networks or the services provided by Virtual Networks. Trust computation Model Based on Improved Bayesian is proposed by Miao Zhang in [17]. G Athanasiou et al. in [18] use fuzzy logic for trust calculations. Binomial opinion and subjective modelling are used for trust score calculation in [5]. A machine learning (ML)-based trust calculation model was proposed by Jayasinghe U et al. in [19], which enhances the capability of the trust evaluation technology to recognise trusted interactions. The trust evaluation method is more credible when it uses the K-means clustering technique as presented in [20] since it can effectively filter out malicious devices. G Athanasiou et al. in [21] use fuzzy logic for trust calculations. The calculation of the trust score is determined to be most effective when using machine learning (ML) approaches since data can be processed effectively and with high computational efficiency [4], [22]. In this research, we present a trust score computation model that uses a weighted average approach.

A. Different approaches for Trust Score Calculation

1) *Evidence-based*: This method uses evidence to support the trustworthiness of an entity. For example, an entity that has provided verifiable proof of its identity, such as a digital certificate, is supposed to have a higher trust score than one that has not [11].

2) *Behaviour-based*: This approach determines a device or user’s trust score based on how they behave on the network. For instance, a device or user may receive a higher trust score if they frequently carry out acts that are considered to be trustworthy [15].

3) *Collaboration-based*: This approach calculates the trust score based on how well one entity collaborates and interacts with other entities in the network [14]. For instance, an entity with a high trust score may frequently engage in successful transactions with other trusted entities.

4) *Peer review based*: In this technique, other network users or devices evaluate a device’s or user’s trustworthiness and assign a trust score based on their judgement [14].

5) *Risk assessment*: In this approach, the entity under consideration is subjected to a thorough risk assessment that considers factors like the possibility that a security event will occur and the potential effects such an incident may have on the organization. The outcome of the risk analysis can be utilized to determine the entity's trust score [13].

6) *Statistical analysis*: This method involves using statistical analysis techniques to evaluate the trustworthiness of an entity. In order to do this, it may be necessary to analyze large volumes of data to find trends and patterns that are suggestive of reliability, and then use this data to determine the entity's trust score [16], [17].

7) *Machine learning-based*: These techniques use machine learning algorithms to study the behaviour and interactions of entities in the network to determine trust scores. This can include developing a model based on past data to predict the reliability of new entities in the network [19]–[21].

8) *Hybrid methods*: These methods combine two or more of the above methods to calculate trust scores.

While many works have utilized machine learning-based trust evaluation models, for a number of reasons, we have opted to use a weighted average-based approach. Firstly, using a weighted average evaluation allows us to explicitly state the elements and their associated weights, guaranteeing that the process of determining trustworthiness is transparent. Secondly, the system can operate in a generic environment and weights allocated to various parameters can be decided in accordance with organisations' or individuals' requirements and priorities, giving users more control over the assessment criteria. Also, if there is a need to modify the evaluation criteria or adjust the relative importance of factors, the weights can be updated accordingly and this is not easily possible in an ML-based trust evaluation technique. Lastly, the weighted average approach's effectiveness and simplicity allow us to deploy and maintain the trust evaluation system without the need for a lot of data and time to train and test the system. This method is computationally less intensive compared to complex ML models.

III. PROPOSED TRUST CALCULATION APPROACH

Trust score is a dynamic score determined based on a user's or a device's behaviour and actions. It aids in our comprehension of how much a person, or a device can be trusted. There will be a check on malicious activities in the network since anomalous behaviour of the user or device will be reflected in the estimated trust score.

The proposed trust evaluation system is designed to be transparent, customizable, and user-friendly. It utilizes various trust-related parameters such as logs, device details, access requests, user details etc., to calculate the trust score [23].

Based on these parameters, the algorithm uses a weighted average method to determine the trust score. Each characteristic is given a weight that corresponds to its relative significance in figuring out trustworthiness. This enables organisations or individuals to customise the evaluation process to suit

their particular requirements and objectives. The method promotes openness and enables users to understand how their trustworthiness is calculated by providing explicit visibility into the elements considered and their corresponding weights. With its customizable nature, the system offers an easy-to-use solution for assessing trust, empowering users to make informed decisions.

The evaluated trust score is forwarded to the Policy Decision Point of Zero Trust system where a decision is made on whether to grant/deny the request made by the user/device based on the trust score

Table I indicates a few potential parameters for evaluating trust and how they might affect trust score in Zero Trust environment:

IV. SAMPLE SCENARIO FOR TRUST SCORE CALCULATION

Trust evaluation is implemented using a trust algorithm. The user, user attributes or roles, and device information like IP address, OS, browser used, access time, etc. are all inputs used by the trust algorithm. The inputs are taken from the device identification database and access logs. Using these inputs, the Trust Score is calculated [24] which is a numerical value between 0 and 10. The calculated trust score is used by the Policy Decision Point (PDP) to determine whether or not access to the resource should be permitted. The trust algorithm calculates trust scores using a weighted average method, where each input parameter is assigned a weight and the weights vary depending on the on importance of each parameter type. The following formula is used to determine trust score:

$$\text{TrustScore}(TS) = \frac{p1 * w1 + p2 * w2 + \dots + pn * wn}{p1 + p2 + \dots + pn} \quad (1)$$

Where 'p1', 'p2', .. 'pn' are the selected parameters and 'w1', 'w2', ... , 'wn' are the weights assigned for the parameters.

The Table II shows a sample trust score calculation using only four parameters i.e, Time of access, Device, OS and Browser used for accessing the resource. More parameters will be considered in the actual scenario. The threshold value of the trust score is taken as 6, above which access is allowed and below 6, access is denied.

It is assumed that the system under consideration is an internal organizational setup, meaning that it operates within the confines of the organization's network and infrastructure. Another assumption is that device details utilized in the trust evaluation process are obtained from the asset database of the organization. It is also assumed that internal authentication and authorization mechanisms of the organization are in place.

Here, all devices owned and managed by the organization are grouped as 'managed' and devices with anonymous IP are considered as 'unmanaged'. Applications' CVSS scores are retrieved from the CVE database [25]. The greater the vulnerability, the higher the CVSS score. Since a greater trust score indicates less vulnerability, the CVSS score is converted to 'Weight' in the trust score by deducting it from 10.

In the considered scenario, we have configured a high trust score (10) for resource access from managed devices and

TABLE I
PARAMETERS AND THEIR IMPACT ON TRUST SCORE

Parameters	Description	Impact on Trust Score
IP address	IP of the client requesting access to resources	In comparison to blacklisted IPs, non-blacklisted IPs will have a greater trust score.
User name and role	User name and role of the user	The trust score will be zero and access will be prohibited if the username is incorrect. Role-based trust scores can also be calculated such that someone in a more senior position would receive a higher trust score and would therefore have access to critical resources.
Location	Location from which the access is requested. The location can be roughly estimated using IP.	The user's trust score will drop if a device is found to have been moved inappropriately (for example, if they log in from two distant cities in a short time that is not acceptable for a flight).
Time	Time of the request	Request at an odd time/non-office hours can lead to a reduced trust score.
Device details	Device details of the client (OS version, Browser version etc.)	A device's trust score will drop if it suddenly changes its OS or logs in using a new browser, both of which can be signs of an attack.
Device health	Software and firmware updates	Device health can be monitored periodically and trust score can be reduced if the latest patches are not installed or software updates are not done.
History of the user and device	History derived from the previous event logs	A user's or a device's excessive number of unsuccessful login attempts may indicate an attack, and in such cases, the trust score will be lowered.

a low trust score (0) for resource access from unmanaged devices. Additionally, under this experimental situation, high weightage is given to resource access during office hours(10) and a reduced weight is given to accesses outside office hours (8). CVSS scores associated with vulnerabilities of the used version of browsers and OSs are also depicted in the table. Algorithm 1 shows how the Decision comes up after Trust Score Evaluation in Table II

Algorithm 1 Sample Trust Score Evaluation Scenario with four parameters

```

if (Managed Device) then
  if Access Request During Office hours then
    if Vulnerable Application then
      Low Trust Score
      Access Denied
    else
      High Trust Score
      Access Allowed
    end
  end
  if Access Request During Non-Office hours then
    Low Trust Score
    Access Denied
  end
end
if (Un-Managed Device) then
  Low Trust Score
  Access Denied
end

```

Consider the third scenario in the Table II. The OS vulnerability which has a CVSS score of 10, and consequently an 'OS weight' of 0, prevents access from being granted. Low OS weight causes the computed Trust Score to be below the threshold.

From the table, we can understand that vulnerabilities in applications and access during non-office hours have led to a reduction in trust scores. A managed and fully patched device

will have a high trust score value.

A. Using Threat Intelligence Data for Trust Evaluation

Using threat intelligence data for trust evaluation can improve the accuracy and effectiveness of the evaluation procedure. Threat intelligence involves collecting and analyzing information about potential threats, vulnerabilities, and malicious activities in the digital world. By incorporating threat intelligence data into trust evaluation, organizations can determine the trustworthiness of individuals or entities based on their exposure to security risks. This information may include signs of compromised devices, known malicious behaviours, suspicious IP addresses, patterns of cyber attacks, etc.

Tools like MISP(Malware Information Sharing Platform) [26] allow users to query and search its database for specific IP addresses or utilize its built-in correlation features to identify IP addresses that have been reported as blacklisted by trusted sources. Once the IP address is obtained from the access request, using APIs, we can query if the IP is blacklisted or not. Blacklisted IPs can be given a '0' trust score.

B. A web server resource access implementation using the derived trust score

The web server resource access using the Zero Trust Access is depicted in Figure 3.

As shown in the diagram, the resources deployed in the web server are accessed by the user using a web browser. To access this resource, the user may use managed devices or unmanaged devices. Managed devices are owned and maintained by the organization. Unmanaged device is the user's personal device such as a mobile with an unknown IP. We have configured the policy to allow resource access if the calculated trust score is greater than or equal to 6. For this experiment, we have configured a high weightage (10) for the resource access from managed devices and a low weightage (0) for the resource access from unmanaged devices. Hence, as per the defined

TABLE II
SAMPLE TRUST SCORE CALCULATION

Time			Device		OS			Browser			TRUST SCORE	DECISION
Sl. No	p1	w1	p2	w2	p3	CVSS	w3	p4	CVSS	w4		
1	Office Hours	10	Managed	10	Linux (Kernel 6.0)	6.4 (CVE-2022-44034)	3.6	Firefox (version 61.0.1)	5.8 (CVE-2018-12386)	4.2	6.95	ACCESS ALLOWED
2	Office Hours	10	Managed	10	Windows (Version 8.1)	8.1 (CVE-2022-41081)	1.9	Chrome (version 60.0)	7.2 (CVE-2017-15404)	2.8	6.175	ACCESS ALLOWED
3	Office Hours	10	Managed	10	Kali Linux (Kernel 4.2.2)	10 (CVE-2012-6392)	0	Chrome (version 60.0)	7.2 (CVE-2017-15404)	2.8	5.7	ACCESS DENIED
4	Non-Office Hours	8	Managed	10	Linux (Kernel 6.0)	6.4 (CVE-2022-44034)	3.6	Firefox (version 61.0.1)	5.8 (CVE-2018-12386)	4.2	6.45	ACCESS ALLOWED
5	Office Hours	10	Unmanaged	0	Windows (Version 8.1)	8.1 (CVE-2022-41081)	1.9	Chrome (version 60.0)	7.2 (CVE-2017-15404)	2.8	3.61	ACCESS DENIED
6	Non-Office Hours	8	Managed	10	Windows (Version 8.1)	8.1 (CVE-2022-41081)	1.9	Chrome (version 60.0)	7.2 (CVE-2017-15404)	2.8	5.675	ACCESS DENIED
7	Non-Office Hours	8	Unmanaged	0	Linux (Kernel 6.0)	6.4 (CVE-2022-44034)	3.6	Chrome (version 60.0)	7.2 (CVE-2017-15404)	2.8	3.6	ACCESS DENIED
8	Non-Office Hours	8	Unmanaged	0	Windows (Version 8.1)	8.1 (CVE-2022-41081)	1.9	Firefox (version 61.0.1)	5.8 (CVE-2018-12386)	4.2	3.48	ACCESS DENIED

policy, access from any unmanaged device will reduce the trust score.

We have installed and configured an Apache2 web server as the resource. The users can access the resources available on the web server using any browser application. From the HTTP request, we gather details about the device's IP address, operating system, browser, and other information. We can also determine whether the device is managed or not. The request is processed by the Identity Access Management (IAM), Threat Intelligence, Trust Score Calculation, and Policy Decision Point modules (PDP). PDP decides whether to grant or refuse access to the web server. Un-trusted entities have been redirected to an error page using a reverse proxy configured using the 'Nginx'.

The Asset Database's device information is mapped to the IP address from the request. Asset Database offers details such as open ports and MAC addresses. The asset database is continuously updated by the CDM module which continuously scans the network and stores information such as IP, MAC, OS, and open ports of every device in the network. These details are fed into the threat intelligence module. The CVE database is used to retrieve information about vulnerabilities of applications and the associated CVSS score. Threat information can also be obtained from tools like MISP [26], STIX/TAXII [27] etc. The trust score calculation module receives these details and assigns the user a trust score using the information obtained from threat intelligence.

At PDP, the calculated trust score value is compared to a predetermined threshold value; if the value is higher than the

threshold, access to the web server is permitted; if it is lower than the threshold, access to the web server is prohibited and those users are redirected away from the web server using a reverse proxy.

For instance, we depicted three cases in the figure where three users requested server access. 'User 1' has a managed and updated device, 'User 2' has a managed device with a vulnerable application, and 'User 3' has an unmanaged device. User 2 will have a lower trust score because of the vulnerability of his application. For example, consider User 2 has an outdated browser (Google Chrome version 3.0.193.2). As per the CVE, this version of the browser is highly vulnerable (CVE-2009-3932) and has a very high CVSS score (9.3/10). Hence, the access request from 'User 2' is denied even though he uses a managed device. Whereas, as per the CVE, the Chrome browser with version 96.0.4664.45 has less vulnerability with a CVSS score of 4.3/10 (CVE-2021-38021). However, due to his unmanaged device, requests from User 3 will have a lower trust score and access will not be allowed.

As a result, only User 1 with a trust score higher than the required level will be granted access to the web server.

V. CONCLUSION

The zero-trust framework outlines a set of principles to eliminate inherent trust and provide security through continuous user and device verification. An essential part of ZTA is trust evaluation and the suggested approach makes use of trust-related attributes, employs a weighted average method, and prioritizes transparency, adaptability, and usability to provide

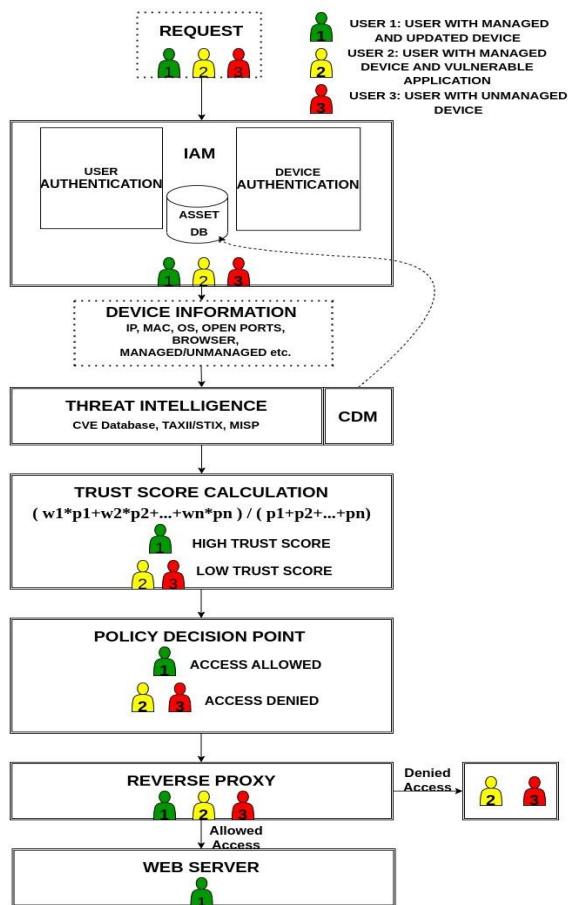


Fig. 3. A web server resource access implementation using Zero Trust Access

an extensive framework for evaluating trust. This method is easier to interpret and understand, does not rely heavily on vast amounts of training data, and allows for direct control over the evaluation process. These factors make the weighted average approach a superior choice for trust evaluation in zero-trust environments, offering a practical and effective alternative to ML-based models.

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